2023 WSMO Accuracy Round Solutions

SMO Team

AR 1: Let $x = \sqrt{69 + \sqrt{69 + \sqrt{69 \ldots}}}$. Find the value of $(2x-1)^2$.

Answer: $\boxed{277}$

Solution: We have

$$\begin{aligned} x &= \sqrt{69+x} \implies \\ x^2 &= 69 + x \implies \\ x^2 - x - 69 &= 0 \implies \\ x &= \frac{1 \pm \sqrt{277}}{2} \implies \\ (2x-1)^2 &= \left(2\left(\frac{1 \pm \sqrt{277}}{2}\right) - 1\right)^2 \\ &= \left(\pm\sqrt{277}\right)^2 = \boxed{277}. \end{aligned}$$

AR 2: When Bob is in precalculus, he gets bored and writes all the permutations in "precal". Since he is not very smart, it takes him 5 seconds to write each permutation. When Bob advances to calculus, he gets bored and writes all the permutations in "calculus". He is smart and can now write each permutation in 2 seconds. Find the positive difference in minutes between the time it takes for him to write the permutations of "precal" and "calculus".

Answer: $\boxed{108}$

Solution: There are 6! permutations of "precal", meaning it takes Bob

$$6! \cdot 5 = 720 \cdot 5 = 3600 \text{ seconds} \implies \frac{3600}{60} = 60 \text{ minutes}$$

to write all the permutations. There are $\frac{8!}{2!2!2!} = 5040$ permutations of "calculus", meaning it takes Bob

$$5040 \cdot 2 = 10080 \text{ seconds} \implies \frac{10080}{60} = 168 \text{ minutes}$$

to write all the permutations. Thus, our answer is

$$168 - 60 = \boxed{108}.$$

AR 3: $f(x) = x^3 - 8x^2 + 10x - 4$ has complex roots a, b, and c. Denote $P(n) = a^n + b^n + c^n$. Find $P(-1)P(0)P(1)$.

Answer: $\boxed{60}$

Solution: We have

$$\begin{aligned} P(-1)P(0)P(1) &= \left(a^{-1}+b^{-1}+c^{-1}\right)\left(a^0+b^0+c^0\right)\left(a^1+b^1+c^1\right) \\ &= \left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)(1+1+1)\,(a+b+c) \\ &= \frac{3(ab+ac+bc)(a+b+c)}{abc} \\ &= \frac{3(10)(8)}{4} = \boxed{60}. \end{aligned}$$

AR 4: Ten people, including Bob and his 3 friends, stand in a line in random order. Given that Bob is not at either end, the probability that the people directly in front of and behind Bob are both his friends can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{13}$

Solution: The probability that the person in front of Bob is his friend is $\frac{3}{9}$. The probability that the person behind Bob is his friend given that the person in front of Bob is his friend is $\frac{2}{8}$. So, our answer is

$$\left(\frac{3}{9}\right)\left(\frac{2}{8}\right) = \frac{6}{72} = \frac{1}{12} \implies 1+12 = \boxed{13}.$$

AR 5: Bob flips $x+1$ coins and Bobby flips x coins, where x is an integer chosen uniformly at random from $[27, 100]$. The expected probability that Bob gets more heads than Bobby can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{3}$

Solution: We will count the number of outcomes in which Bob and Bobby flip coins such that Bob gets more heads than Bobby, and divide that by the total number of possible outcomes, $2^{x+1} \cdot 2^x = 2^{2x+1}$. Consider Bob's and Bobby's first x flips. There are a total of $2^x \cdot 2^x = 2^{2x}$ possible outcomes. Among these,

$$\sum_{i=0}^{x} \binom{x}{i}\binom{x}{i} = \sum_{i=0}^{x} \binom{x}{i}\binom{x}{x-i} = \binom{2x}{x}$$

are the outcomes in which Bob and Bobby have the same number of heads. Now, we consider Bob's last coin flip. For these $\binom{2x}{x}$ outcomes, Bob must flip a head to end up with more heads than Bobby. For the remaining $2^{2x} - \binom{2x}{x}$ outcomes, in exactly half of them Bob already has more heads than Bobby, and in the other half, Bobby has more. If Bob already has more heads, then regardless of the final flip, he will still have more heads. If he has fewer heads, then the final flip will not change that. So, the number of favorable outcomes is

$$\binom{2x}{x} + \frac{2^{2x} - \binom{2x}{x}}{2} \cdot 2 = 2^{2x}.$$

Therefore, the probability that Bob ends up with more heads than Bobby is

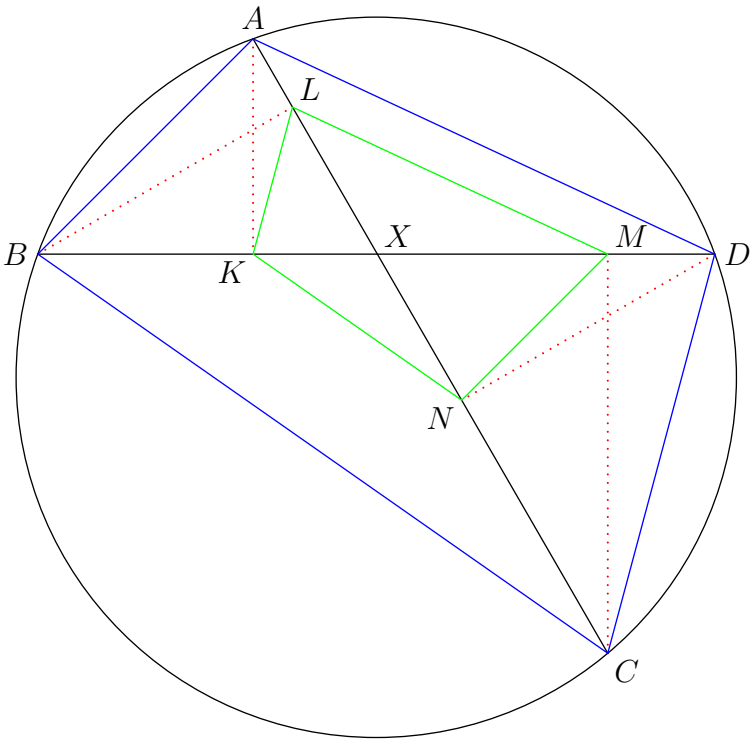
$$\frac{2^{2x}}{2^{2x+1}} = \frac{1}{2}.$$

So, the expected probability over x from $[27, 100]$ is $\frac{1}{2} \implies 1 + 2 = \boxed{3}$.

AR 6: In quadrilateral $ABCD$, there exists a point O such that $AO = BO = CO = DO$ and $\angle(AOB) + \angle(COD) = 120°$. Let K, L, M, N be the feet of the perpendiculars from A to BD, B to AC, C to BD, and D to AC. If $[ABCD] = 20$, find $([KLMN])^2$.

Answer: $\boxed{25}$

Solution:



The existence of point O implies that $ABCD$ is a cyclic quadrilateral. Now, we have

$$\begin{aligned}
\angle AXB &= 180° - \angle AXD \\
&= 180° - (180° - \angle XAD - \angle XDA) \\
&= \angle XAD + \angle XDA \\
&= \angle CAD + \angle BDA \\
&= \frac{\overset{\frown}{CD}}{2} + \frac{\overset{\frown}{AB}}{2} \\
&= \frac{\angle COD}{2} + \frac{\angle AOB}{2} \\
&= \frac{120°}{2} = 60°.
\end{aligned}$$

So,

$$\begin{aligned} KM &= KX + XM \\ &= AX\cos 60^\circ + XC\cos 60^\circ \\ &= \frac{AX}{2} + \frac{XC}{2} \\ &= \frac{AC}{2}. \end{aligned}$$

In the same manner, we have $LN = \frac{BD}{2}$. We have

$$\begin{aligned} [KLMN] &= \frac{1}{2}(KM)(LN)\sin\angle LXK \\ &= \frac{1}{2}\left(\frac{AC}{2}\right)\left(\frac{BD}{2}\right)\sin\angle AXB \\ &= \frac{1}{4}\left(\frac{1}{2}(AC)(BD)\sin\angle AXB\right) \\ &= \frac{[ABCD]}{4} \implies \\ [KLMN] &= \frac{20}{4} = 5 \implies \\ [KLMN]^2 &= 5^2 = \boxed{25}. \end{aligned}$$

AR 7: How many ordered triplets of integers (a, b, c) satisfy $a^2 + 2ab + b^2 = c^2 - 6c + 9$ and $-2 \leq a, b, c \leq 7$?

Answer: $\boxed{95}$

Solution: Taking the square root, we have

$$|a + b| = |c - 3| = d.$$

For each of the 9 values of $c \neq 3$, there are 10 ordered pairs (a, b) such that $|a + b| = d \neq 0$. For $c = 3$, there are only 5 ordered pairs (a, b) such that $|a + b| = d = 0$. So, there are a total of

$$9 \cdot 10 + 5 = \boxed{95}$$

such ordered triplets.

AR 8: Let $f(x) = x^3 - 3x^2 + 4x + 5$ have complex roots a, b, and c. Then,

$$\frac{1}{a^2 + b^2} + \frac{1}{a^2 + c^2} + \frac{1}{b^2 + c^2}$$

can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{68}$

Solution: Note that

$$a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + ac + bc) = 3^2 - 2 \cdot 4 = 1.$$

So,

$$\frac{1}{a^2+b^2}+\frac{1}{a^2+c^2}+\frac{1}{b^2+c^2}=\frac{1}{1-c^2}+\frac{1}{1-b^2}+\frac{1}{1-a^2}=\frac{1}{a_1}+\frac{1}{b_1}+\frac{1}{c_1}$$

where

$$a_1=1-a^2,\quad b_1=1-b^2,\quad c_1=1-c^2.$$

Note that a_1, b_1, c_1 are the roots of $f(\sqrt{1-x})$. Solving, we have

$$\begin{aligned}\left(\sqrt{1-x}\right)^3-3\left(\sqrt{1-x}\right)^2+4\left(\sqrt{1-x}\right)+5&=0\\(1-x)\sqrt{1-x}+4\sqrt{1-x}&=3(1-x)-5\\(5-x)\sqrt{1-x}&=-3x-2\\(5-x)^2(1-x)&=(-3x-2)^2\\-x^3+11x^2-35x+25&=9x^2+12x+4\\x^3-2x^2+47x-21&=0.\end{aligned}$$

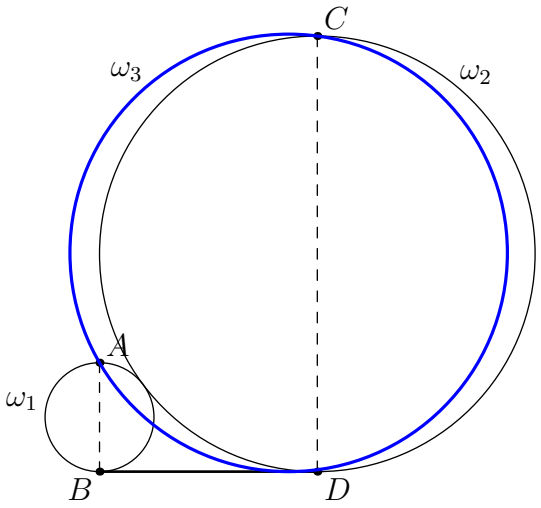
So,

$$\frac{1}{a_1}+\frac{1}{b_1}+\frac{1}{c_1}=\frac{b_1c_1+a_1c_1+a_1b_1}{a_1b_1c_1}=\frac{47}{21}\implies 47+21=\boxed{68}.$$

AR 9: Circles ω_1 and ω_2 with radii 1 and 4, respectively, are externally tangent to each other. The diameters of ω_1 and ω_2 are AB and CD, respectively, such that $AB \parallel CD$ and BD is a common external tangent to the circles. A third circle ω_3 passes through A and C and is tangent to BD. The minimum possible value of the radius of ω_3 can be expressed as $\frac{a-b\sqrt{c}}{d}$, where a, b, c, and d are positive integers, c is not divisible by the square of any prime, and $\gcd(a,b,d)=1$. Find $a+b+c+d$.

Answer: $\boxed{95}$

Solution:



We have

$$BD=\sqrt{(4+1)^2-(4-1)^2}=4.$$

Let $B=(0,0)$ and $D=(4,0)$. This means $A=(0,2)$ and $C=(4,8)$. Suppose the center of ω_3 is (x,y). Since ω_3 is tangent to BD, the radius of ω_3 must be y. Since A, C lie on ω_3, we have

$$x^2+(y-2)^2=(x-4)^2+(y-8)^2=y^2.$$

From

$$x^2 + (y-2)^2 = (x-4)^2 + (y-8)^2$$

we have

$$8x + 12y = 76 \implies x = \frac{19}{2} - \frac{3}{2}y.$$

Substituting into

$$x^2 + (y-2)^2 = y^2$$

we have

$$\frac{9}{4}y^2 - \frac{65}{2}y + \frac{377}{4} = 0 \implies y = \frac{65 - 8\sqrt{13}}{9} \implies 65 + 8 + 13 + 9 = \boxed{95}.$$

AR 10: In regular tetrahedron T of side length 12, let S_1 be the sphere inscribed in T and let S_2 be the sphere circumscribed around T. Let R be a rectangular prism such that all points on S_1 lie strictly inside or are touching R and all points on R lie strictly inside or are touching S_2. The minimum possible volume of R can be expressed as $m\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m + n$.

Answer: $\boxed{54}$

Solution: The height of T is $4\sqrt{6}$, and the inradius is a fourth of that, which is $\sqrt{6}$. This means each dimension of R has to be at least $2\sqrt{6}$. Take a cube of side $2\sqrt{6}$ concentric with S_1. This cube completely contains S_1 and lies inside S_2. Thus, our answer is

$$2\sqrt{6} \cdot 2\sqrt{6} \cdot 2\sqrt{6} = 48\sqrt{6} \implies 48 + 6 = \boxed{54}.$$